FIT3080 Artificial Intelligence
Practice Questions in Knowledge Representation

1. [**Entailment**] Prove the following lemma: $\alpha \models \beta$ iff $\alpha \wedge \neg\beta$ is unsatisfiable.

2. [**Propositional logic**] Say whether each of the following sentences is syntactically legal in Propositional Logic, and if not, why:

   - $P \wedge \neg P$
   - $\neg(Q \vee R)\neg Q \Rightarrow \neg\neg P$

3. [**Forward/backward chaining**] Apply forward and backward reasoning to the following horn clauses to prove PassExam.
   R1: $Enroll \wedge StudyHard \wedge Understand \Rightarrow PassExam$
   R2: $AskGoodQuestions \Rightarrow Understand$
   R3: $PayTuitionFee \Rightarrow Enroll$
   R4: $NeverPlayGame \wedge SitAllDay \Rightarrow ReadAndWork$
   R5: $ReadAndWork \Rightarrow StudyHard$
   R6: $StudyHard \Rightarrow AskGoodQuestions$
   $PayTuitionFee$
   $NeverPlayGame$
   $SitAllDay$

4. [**First Order logic**] Using the predicates PERSON and HAS-PET, represent the following sentence in First Order Logic:
   Some people don't have a pet.

5. [**First Order logic**] Assuming predicates $PARENT(p, q)$ and $FEMALE(p)$ and constants $Joan$ and $Kevin$, with the obvious meanings, express each of the following sentences in First-Order Logic. (You may use the abbreviation $\exists^1$ to mean "there exists exactly one.")

   (a) Joan has a daughter (possibly more than one, and possibly sons as well).

   (b) Joan has exactly one daughter (but may have sons as well).

   (c) Joan has exactly one child, a daughter.

   (d) Joan and Kevin have exactly one child together.

   (e) Joan has at least one child with Kevin, and no children with anyone else.

6. [**First Order logic**] Consider a vocabulary with the following symbols:
   $OCC(p, o)$: Predicate. Person p has occupation o.
   $CUST(p1, p2)$: Predicate. Person $p1$ is a customer of person $p2$.
   $BOSS(p1, p2)$: Predicate. Person $p1$ is a boss of person $p2$.
   $Doctor, Surgeon, Lawyer, Actor$: Constants denoting occupation.
   $Emily, Joe$: Constants denoting people.

   Use these symbols to write the following assertions in First Order Logic:

   (a) Emily is a surgeon or a lawyer.

(b) Joe is an actor, but he also holds another job.

(c) All surgeons are doctors.

(d) Joe does not have a lawyer (i.e., is not a customer of any lawyer).

(e) Emily has a boss who is a lawyer.

(f) There exists a lawyer all of whose customers are doctors.

(g) Every surgeon has a lawyer.

7. [**First Order logic**] Indicate which of the following Predicate Calculus sentences are correct representations of the corresponding English sentences. If not, explain why not.

1	$[\forall x$ PERSON$(x)] \Rightarrow [\exists y$ MOTHER$(y,x)]$ Everybody has a mother
2	OLD(DOG(Fido)) Fido is an old dog
3	DOG(Fido) $\wedge$ OLD(Fido) Fido is an old dog
4	$\forall x$ [MATHEMATICAL-THEORY$(x) \Rightarrow x$] All mathematical theories are true
5	$\exists s$ [ARISTOTLE-SAID(s) $\wedge \neg$TRUE(s)] Aristotle told a lie
6	$\neg\exists v\ \exists b$ [BUTCHER$(b) \wedge$ VEGETARIAN(v)] There are no vegetarian butchers
7	$\neg\exists b\ \exists d$ BUTCHER$(b) \wedge$ DOG$(d) \wedge$ OWNS(b,d) No butcher owns a dog

8. [**Substitution**] Consider the following substitutions:
$s1 = \{y|x, z|A\}; s2 = \{x|C, z|y\}$
Perform the composition $s1s2$ and $s2s1$.

9. [**Unification**] For each of the following pairs of literals indicate whether or not they unify. If they do unify, show a most general unifier. If they do not unify, explain why not.

   (a) STRONGER(x, LexLuthor), STRONGER(LoisLane,y)

   (b) HOLDING(x, Girlfriend(x)), HOLDING(Superman,y)

   (c) FLY(Girlfriend(x)), $\neg$FLY(LoisLane)

   (d) $\neg$LIKES(x,Enemy(x)), $\neg$LIKES(Enemy(y), y)

10. [**Unification**] For each pair of atomic sentences, give the most general unifier if it exists. If they do not unify, explain why not.

   (a) $P(A, B, B), P(x, y, z)$.

   (b) $Q(y, G(A, B)), Q(G(x, x), y)$.

   (c) $OLDER(Father(y), y), OLDER(Father(x), John)$.

   (d) $KNOWS(Father(y), y), KNOWS(x, x)$.

11. [**Resolution**] Resolve the following clauses, indicating clearly the literals you are resolving, their *mgu* and the resolvent:
$P(x, y) \vee \neg Q(A)$ and $W(x) \vee Q(x)$.

12. [**Resolution**]Consider the following statements:

    1. `All hounds howl at night.`
    2. `Anyone who has any cats will not have any mice.`
    3. `Light sleepers do not have anything which howls at night.`
    4. `John has a cat or a hound.`
    5. `If John is a light sleeper, then John does not have any mice.`

    (a) Using only the predicates $HOUND$, $HOWL$, $HAVE$, CAT, $MOUSE$ and $LIGHTSLEEPER$, represent these statements as well formed formulas (wffs).

    (b) Convert Statements 1-4 to clauses.

    (c) Use resolution to prove Statement 5.

13. [**Resolution**] From "Horses are animals", it follows that "The head of a horse is the head of an animal." Demonstrate that this inference is valid by carrying out the following steps:

    (a) Translate the premise and the conclusion into a well formed formula (wff) in First Order Logic. Use three predicates: $HEADOF(h, x)$ (meaning "h is the head of x"), $HORSE(x)$, and $ANIMAL(x)$.

    (b) Negate the conclusion, and convert the premise and the negated conclusion into CNF.

    (c) Use resolution to show that the conclusion follows from the premise.